6.6 Theorems on limits of sequences

We can now continue the analysis of sequences, which we began in Section 5.2. To simplify the presentation, let us agree that we say that a sequence $\{a_n\}_{n\in\mathbb{N}}$ satisfies a property **for all large** n if there exists $N \in \mathbb{N}$ such that for all $n > N$ the sequence satisfies this property. The results we obtained for functions all carry over to sequences, so we can state the following 'big' theorem:

Theorem 6.15:

1. The limit of a sequence (if exists) is unique.
2. A convergent sequence is bounded.
3. A sequence that is monotone for all large n cannot be indeterminate.
4. For sequences $\{a_n\}_{n\in\mathbb{N}}, \{b_n\}_{n\in\mathbb{N}}$, if $a_n \le b_n$ for all large n, then $\lim_{n\to\infty} a_n \le \lim_{n\to\infty} b_n$ (if both limits exist).
5. For sequences $\{a_n\}_{n\in\mathbb{N}}, \{b_n\}_{n\in\mathbb{N}}$:
   (a) (Squeeze Theorem) if $\{c_n\}_{n\in\mathbb{N}}$ is a third sequence such that, for all large n, $a_n \le b_n \le c_n$, and $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n$, then b_n has a limit and it is the same limit,
   (b) (Squeeze to $\pm\infty$ Theorem) if $a_n \le b_n$ for all large n, then $\lim_{n\to\infty} a_n = +\infty$ implies $\lim_{n\to\infty} b_n = +\infty$, and $\lim_{n\to\infty} b_n = -\infty$ implies $\lim_{n\to\infty} a_n = -\infty$.
6. If two sequences $\{a_n\}_{n\in\mathbb{N}}, \{b_n\}_{n\in\mathbb{N}}$ have limits $\lim_{n\to\infty} a_n = \ell_a$ and $\lim_{n\to\infty} b_n = \ell_b$, then
   (a) $\lim_{n\to\infty}(a_n \pm b_n) = \ell_a \pm \ell_b$
   (b) $\lim_{n\to\infty}(a_n \cdot b_n) = \ell_a \cdot \ell_b$
   (c) $\lim_{n\to\infty}\left(\frac{a_n}{b_n}\right) = \frac{\ell_a}{\ell_b}$, if for all large n, $b_n \neq 0$,

   whenever the expressions on the right hand side are meaningful.
7. If $\{a_n\}_{n\in\mathbb{N}}$ has the limit ℓ and $g : \mathbb{R} \to \mathbb{R}$ is defined in a deleted neighborhood of ℓ, then
   (a) if $\ell \in \mathbb{R}$ and g is continuous at ℓ, then $\lim_{n\to\infty} g(a_n) = g(\ell)$,
   (b) if $\ell \in \mathbb{R}$, $\lim_{y\to\ell} g(y)$ exists, and $a_n \neq \ell$ for all large n, then $\lim_{n\to\infty} g(a_n) = \lim_{y\to\ell} g(y)$,
   (c) if $\ell = \pm\infty$ and $\lim_{y\to\ell} g(y)$ exists, then $\lim_{n\to\infty} g(a_n) = \lim_{y\to\ell} g(y)$.
8. $\lim_{n\to\infty} a_n = 0$ if and only if $\lim_{n\to\infty} |a_n| = 0$.
9. If a sequence $\{a_n\}_{n\in\mathbb{N}}$ is bounded, and another sequence $\{b_n\}_{n\in\mathbb{N}}$ satisfies $\lim_{n\to\infty} b_n = 0$, then $\lim_{n\to\infty}(a_n b_n) = 0$.

Proof. The proof is completely analogous to the various proofs we've seen for functions. We skip it here. □

For sequences there is an additional useful tool, which relies on the fact that sequences are discrete (as opposed to functions on $\mathbb{R}$):

Theorem 6.16 (Ratio Test)**:** Let $\{a_n\}_{n\in\mathbb{N}}$ be a sequence that is positive for all large n (i.e. there exists $N \in \mathbb{N}$ such that for all $n > N$, $a_n > 0$). Assume that the limit $\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = q$ exists (it may be finite or infinite). Then

- if $q < 1$, then $\lim_{n\to\infty} a_n = 0$,
- if $q > 1$, then $\lim_{n\to\infty} a_n = +\infty$,
- if $q = 1$, the test is inconclusive: the sequence may or may not have a limit.

If $a_n < 0$ for all large n, the same conclusions hold, except that for $q > 1$ we get $\lim_{n\to\infty} a_n = -\infty$ (apply the theorem to $-a_n$).

Proof. The proof is simple and we skip it here. □

Example 6.16: Consider a sequence that we have previously seen:

$$a_n = \frac{n!}{n^{100}}.$$

We have

$$\frac{a_{n+1}}{a_n} = \frac{\frac{(n+1)!}{(n+1)^{100}}}{\frac{n!}{n^{100}}} = \frac{n^{100}(n+1)!}{(n+1)^{100}n!} = \underbrace{\left(\frac{n}{n+1}\right)^{100}}_{\to 1} \underbrace{(n+1)}_{\to+\infty},$$

hence

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = +\infty$$

so that the sequence a_n diverges.

6.7 Fundamental limits and indeterminate forms of exponential type

We have seen before that

$$\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e.$$

Now we show the same result for the function $\left(1 + \frac{1}{x}\right)^x$:

Claim: The function $\left(1 + \frac{1}{x}\right)^x$ has limits as $x \to \pm\infty$, and

$$\lim_{x\to\pm\infty} \left(1 + \frac{1}{x}\right)^x = e.$$

Proof. Observe that the function $\left(1 + \frac{1}{x}\right)^x$ is defined when $1 + \frac{1}{x} > 0$ and $x \neq 0$. Hence it is defined when either $x > 0$ or $x < -1$. We prove for the case $x \to +\infty$.

Let $x \geq 1$ and let $n = \lfloor x \rfloor \geq 1$ be the integer part of x. Then

$$n \leq x < n + 1.$$

Hence

$$\begin{gathered}\frac{1}{n+1} < \frac{1}{x} \leq \frac{1}{n} \\ \Downarrow \\ 1+\frac{1}{n+1} < 1+\frac{1}{x} \leq 1+\frac{1}{n} \\ \Downarrow \\ \left(1+\frac{1}{n+1}\right)^n \leq \left(1+\frac{1}{n+1}\right)^x < \left(1+\frac{1}{x}\right)^x \leq \left(1+\frac{1}{n}\right)^x < \left(1+\frac{1}{n}\right)^{n+1}\end{gathered}$$

So we have:

$$\underbrace{\left(1+\frac{1}{n+1}\right)^{n+1}}_{\to e}\underbrace{\left(1+\frac{1}{n+1}\right)^{-1}}_{\to 1} < \left(1+\frac{1}{x}\right)^x < \underbrace{\left(1+\frac{1}{n}\right)^{n}}_{\to e}\underbrace{\left(1+\frac{1}{n}\right)}_{\to 1}$$

So by the Squeeze Theorem,

$$\lim_{x\to+\infty}\left(1+\frac{1}{x}\right)^x = e.$$

The proof in the case $x \to -\infty$ follows similarly, taking caution with signs. □

Observe that by substituting $y = \frac{1}{x}$ we have:

$$\lim_{x\to\pm\infty}\left(1+\frac{1}{x}\right)^x = \lim_{y\to 0}(1+y)^{\frac{1}{y}} = e.$$

Claim:

$$\lim_{x\to 0}\frac{\ln(1+x)}{x} = 1.$$

Proof.

$$\frac{\ln(1+x)}{x} = \frac{1}{x}\ln(1+x) = \ln\left((1+x)^{\frac{1}{x}}\right).$$

Hence, using the continuity of the logarithm, we have:

$$\lim_{x\to 0}\frac{\ln(1+x)}{x} = \ln\underbrace{\left(\lim_{x\to 0}(1+x)^{\frac{1}{x}}\right)}_{=e} = 1$$

where in the last equality we have used the previous remark (above) about $(1+x)^{\frac{1}{x}}$. □

Claim:

$$\lim_{x\to 0}\frac{e^x-1}{x} = 1.$$

Proof. Set $y = e^x - 1$. Then $y \to 0$ as $x \to 0$, $y \neq 0$ for $x \neq 0$, and $x = \ln(1+y)$. Hence, by the previous claim and the Substitution Theorem (Theorem 6.12, case 2),

$$\lim_{x\to 0}\frac{e^x-1}{x} = \lim_{y\to 0}\frac{y}{\ln(1+y)} = \lim_{y\to 0}\frac{1}{\frac{\ln(1+y)}{y}} = 1.$$

□

Useful limits

$$\lim_{x\to 0}\frac{\sin x}{x}=1$$
$$\lim_{x\to 0}\frac{1-\cos x}{x^2}=\frac{1}{2}$$
$$\lim_{x\to\pm\infty}\left(1+\frac{a}{x}\right)^x=e^a,\qquad (a\in\mathbb{R})$$
$$\lim_{x\to 0}\frac{\log_a(1+x)}{x}=\frac{1}{\ln a},\qquad (a>0,\ a\neq 1)$$
$$\lim_{x\to 0}\frac{a^x-1}{x}=\ln a,\qquad (a>0)$$
$$\lim_{x\to 0}\frac{(1+x)^\alpha-1}{x}=\alpha,\qquad (\alpha\in\mathbb{R})$$

Proof. The first limit was computed in Example 5.13 (see also the Squeeze Theorem above). The others follow from it and from the two claims above:

- Multiplying numerator and denominator by $1+\cos x$ and using $1-\cos^2 x=\sin^2 x$,
$$\frac{1-\cos x}{x^2}=\left(\frac{\sin x}{x}\right)^2\frac{1}{1+\cos x}\to 1^2\cdot\frac{1}{2}=\frac{1}{2}.$$
- If $a=0$ there is nothing to prove. If $a\neq 0$, set $y=\frac{x}{a}$, so that $y\to\pm\infty$ as $x\to\pm\infty$ (with the same sign if $a>0$, the opposite sign if $a<0$). Then, using $\lim_{y\to\pm\infty}\left(1+\frac{1}{y}\right)^y=e$ and the continuity of $t\mapsto t^a$ at $t=e$,
$$\left(1+\frac{a}{x}\right)^x=\left[\left(1+\frac{1}{y}\right)^y\right]^a\to e^a.$$
- Since $\log_a(1+x)=\frac{\ln(1+x)}{\ln a}$, we have $\frac{\log_a(1+x)}{x}=\frac{1}{\ln a}\cdot\frac{\ln(1+x)}{x}\to\frac{1}{\ln a}$.
- If $a=1$ both sides are 0. If $a\neq 1$, write $a^x=e^{x\ln a}$ and set $y=x\ln a$, which tends to 0 and is $\neq 0$ for $x\neq 0$. By the Substitution Theorem (case 2),
$$\frac{a^x-1}{x}=\ln a\cdot\frac{e^y-1}{y}\to\ln a\cdot 1=\ln a.$$
- If $\alpha=0$ both sides are 0. If $\alpha\neq 0$, write $(1+x)^\alpha=e^{\alpha\ln(1+x)}$ and set $y=\alpha\ln(1+x)$, which tends to 0 and is $\neq 0$ for $x\neq 0$ (and $x>-1$). By the Substitution Theorem (case 2),
$$\frac{(1+x)^\alpha-1}{x}=\frac{e^y-1}{y}\cdot\frac{\alpha\ln(1+x)}{x}\to 1\cdot\alpha=\alpha.$$

□

Power functions

Limits of powers of functions

Let $h(x) = [f(x)]^{g(x)}$, let $x_0 \in \{-\infty\} \cup \mathbb{R} \cup \{+\infty\}$ and assume that f, g have limits as $x \to x_0$ and that $f > 0$ near x_0. Observe that $h = e^{g \ln f}$. Hence, if $\lim_{x\to x_0} (g(x) \ln f(x))$ exists (finite or infinite), then by the continuity of the exponential and the fact that for continuous functions we can commute the operations of taking the limit and applying the function:

$$\lim_{x\to x_0} \left([f(x)]^{g(x)} \right) = e^{\lim_{x\to x_0} (g(x) \ln f(x))},$$

where we use the conventions $e^{+\infty} = +\infty$ and $e^{-\infty} = 0$ (that is, $\lim_{t\to+\infty} e^t = +\infty$ and $\lim_{t\to-\infty} e^t = 0$).

So we need to study the exponential of $\lim_{x\to x_0} (g(x) \ln f(x))$. This is the limit of the product of two functions. We know that it is problematic if we get $0 \cdot \infty$. Hence we need to investigate thoroughly in these cases:

1. $\lim_{x\to x_0} g(x) = \pm\infty$ and $\lim_{x\to x_0} f(x) = 1$, so that we get 1^∞.
2. $\lim_{x\to x_0} g(x) = 0$ and $\lim_{x\to x_0} f(x) = 0$, so that we get 0^0.
3. $\lim_{x\to x_0} g(x) = 0$ and $\lim_{x\to x_0} f(x) = +\infty$, so that we get ∞^0.

Example 6.17: Determine $\lim_{x\to+\infty} x^{\frac{1}{x}}$.

We see that this has the form ∞^0. Let $y = \frac{1}{x}$, so that the problem becomes $\lim_{y\to0^+} (1/y)^y$. We see that

$$x^{\frac{1}{x}} = \left(\frac{1}{y}\right)^y = e^{y \ln \frac{1}{y}} = e^{-y \ln y}.$$

We will later prove that $\lim_{y\to0^+} (y \ln y) = 0$ (see Example 10.28 with $\alpha = 1$: setting $x = \frac{1}{y}$, we have $-y \ln y = \frac{\ln x}{x} \to 0$ as $x \to +\infty$), so that

$$\lim_{x\to+\infty} x^{\frac{1}{x}} = \lim_{y\to0^+} e^{-y \ln y} = e^{\lim_{y\to0^+} (-y \ln y)} = e^0 = 1.$$

Chapter 7

Series

7.1 Series and their partial sums

In this chapter we study **series**: infinite sums of the form

$$a_0 + a_1 + a_2 + \cdots + a_n + \cdots = ?$$

Making sense of such infinite sums – deciding when they have a value, and what that value is – is one of the fundamental problems of analysis. Series will recur later in these notes, for instance when we connect them to improper integrals (Chapter 13).

Series and their partial sums

Given a sequence $\{a_n\}_{n\in\mathbb{N}}$ of real numbers, we form the **partial sums**

$$s_n = a_0 + a_1 + a_2 + \cdots + a_n = \sum_{k=0}^{n} a_k.$$

The **series** (or infinite sum) associated to $\{a_n\}_{n\in\mathbb{N}}$ is defined as the limit of the partial sums:

$$\sum_{k=0}^{\infty} a_k = \lim_{n\to\infty} s_n = \lim_{n\to\infty} \sum_{k=0}^{n} a_k.$$

If the limit exists and is finite, we say the series *converges*; if the limit is infinite, the series *diverges*; if the limit does not exist, the series is *indeterminate*.

7.2 Geometric series

One of the most important examples is the geometric series.

Geometric series

For $r \in \mathbb{R}$, the geometric series is

$$\sum_{n=0}^{\infty} r^n = 1 + r + r^2 + r^3 + \cdots .$$

Its partial sums are given by the finite geometric sum formula (Chapter 0):

$$s_n = 1 + r + r^2 + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r} \quad \text{for } r \neq 1.$$

Taking the limit as $n \to \infty$, we obtain:

$$\sum_{n=0}^{\infty} r^n = \begin{cases} \frac{1}{1-r} & \text{if } |r| < 1 \quad \text{(the series converges)}, \\ +\infty & \text{if } r \geq 1 \quad \text{(the series diverges)}, \\ \text{does not exist} & \text{if } r \leq -1 \quad \text{(the series is indeterminate)}. \end{cases}$$

(For $r = 1$ the formula for s_n does not apply, but directly $s_n = n + 1 \to +\infty$. For $r = -1$ the partial sums alternate: $s_n = 1, 0, 1, 0, \ldots$)

Zeno's paradoxes

Achilles, the fastest runner of Greek legend, races a tortoise that starts 100 *meters ahead. By the time Achilles reaches the tortoise's starting point, it has crawled a little further; by the time he covers that gap, it has moved on again; and so on forever. There are infinitely many catch-up stages, so it seems Achilles can never overtake the tortoise. A related paradox, the* dichotomy*, says that to reach a wall you must first cover half the distance, then half of what remains, and so on, so you never arrive. The geometric series resolves the mathematical puzzle: infinitely many stages can add up to a finite total. The wall is reached after* $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 1$*, the whole distance. If Achilles runs ten times faster than the tortoise, the catch-up distances are* $100 + 10 + 1 + \frac{1}{10} + \cdots = \frac{100}{1 - \frac{1}{10}} = 111.\overline{1}$ *meters, and after that Achilles is in front.*

These puzzles go back to Zeno of Elea (c. 490–425 BC), from a Greek city in southern Italy. He was a pupil of the philosopher Parmenides, who taught that motion is an illusion, and devised the paradoxes to defend him. His book is lost; we know the paradoxes mainly through Aristotle.

7.3 The Cauchy Condensation Test and p-series

p-Series

For $p > 0$, a **p-series** is the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$. In the case $p = 1$, the series $\sum_{n=1}^{\infty} \frac{1}{n}$ is called the **harmonic series**.

To determine exactly when a p-series converges, we introduce a new comparison tool that is especially well-suited to decreasing sequences.